2026 春《操作系统》课程实验报告

实验 0

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一.环境准备

1.git 环境准备

- 初始化 Git 仓库

```Bash
git init
```

- 创建 `.gitignore`

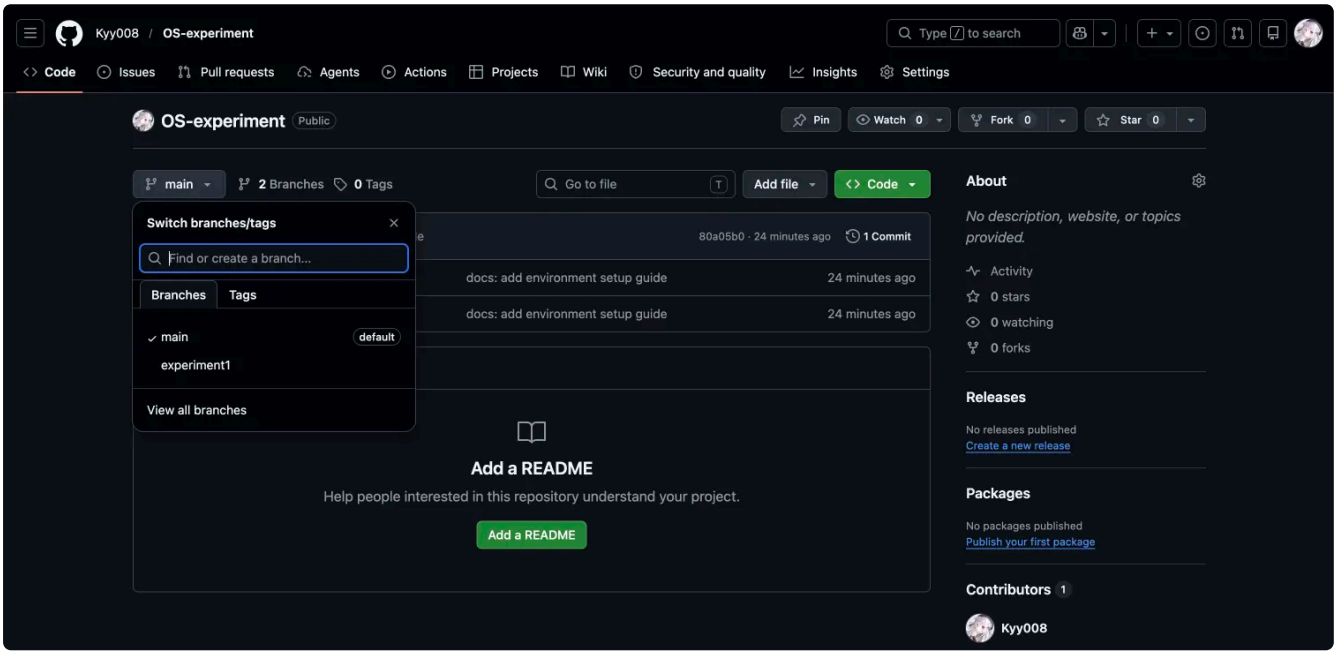
```Bash
touch .gitignore
echo .DS_Store >> .gitignore
```

- 关联远程仓库

```Bash
git remote add origin https://github.com/Kyy008/OS-experiment
```

- 创建并推送实验一分支

```Bash
git branch exp1
git push -u origin exp1
```



- 切换到实验一分支

```bash
git checkout experiment1
```

2.Docker 环境准备

- 拉取 openEuler 镜像

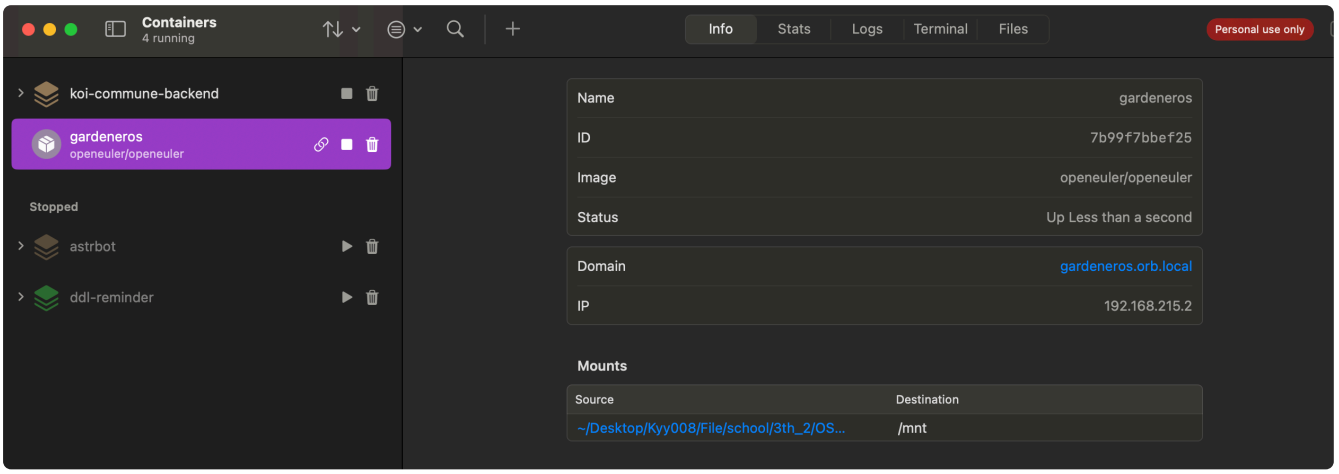
```bash
docker pull openeuler/openeuler
```

```
~/De/K/F/school/3th_2/OS/experiment main ··· 0.015s 11:41:05
└─ docker pull openeuler/openeuler
Using default tag: latest
latest: Pulling from openeuler/openeuler
c2a5d1e5887b: Pull complete
ba0e6af23b31: Pull complete
Digest: sha256:990f5a8528dc3375a358629bcb5500351f433a49922dca3588bcf32ecc5b7022
Status: Downloaded newer image for openeuler/openeuler:latest
docker.io/openeuler/openeuler:latest
```

- 在实验目录下启动容器，把当前的实验目录挂载到容器里的 `/mnt`

Bash

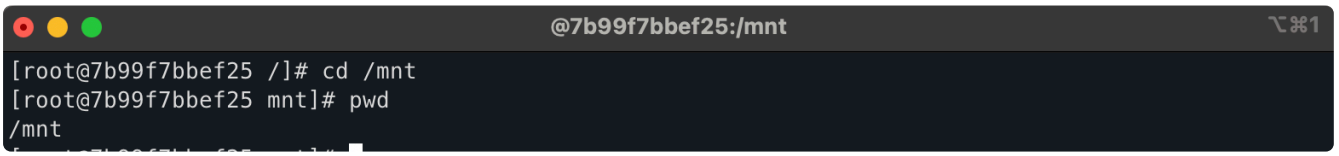
```
docker run -it --name gardeneros \
  --mount type=bind,source="$(pwd)",destination=/mnt \
  openeuler/openeuler /bin/bash
```



- 进入容器内的挂载目录

Bash

```
cd /mnt
pwd
```



- 为容器安装基础工具：

Bash

```
dnf install -y curl vim gcc git
```

```
[root@7b99f7bbef25 mnt]# dnf install -y curl vim gcc git
OS                                                  2.3 MB/s | 2.8 MB     00:01
everything                                          5.4 MB/s |  16 MB     00:02
EPOL                                                5.7 MB/s | 6.3 MB     00:01
debuginfo                                           3.9 MB/s | 4.2 MB     00:01
source                                              1.6 MB/s | 1.7 MB     00:01
update                                              4.9 MB/s |  35 MB     00:07
update-source                                       2.2 MB/s | 1.7 MB     00:00
Package curl-8.4.0-28.oe2403sp2.aarch64 is already installed.
Dependencies resolved.
```

```
  systemtap-sdt-devel-5.0-3.oe2403sp2.aarch64
  util-linux-2.39.1-35.oe2403sp2.aarch64
  vim-common-2:9.0.2092-29.oe2403sp2.aarch64
  vim-enhanced-2:9.0.2092-29.oe2403sp2.aarch64
  vim-filesystem-2:9.0.2092-29.oe2403sp2.noarch
  xkeyboard-config-2.39-4.oe2403sp2.noarch
  zip-3.0-32.oe2403sp2.aarch64

Complete!
[root@7b99f7bbef25 mnt]#
```

3.配置 Rust 开发环境

- 使用官方脚本安装 Rust

Bash

```
curl https://sh.rustup.rs -sSf | sh
```

```
Rust is installed now. Great!

To get started you may need to restart your current shell.
This would reload your PATH environment variable to include
Cargo's bin directory ($HOME/.cargo/bin).

To configure your current shell, you need to source
the corresponding env file under $HOME/.cargo.

This is usually done by running one of the following (note the leading DOT):
. "$HOME/.cargo/env"            # For sh/bash/zsh/ash/dash/pdksh
source "$HOME/.cargo/env.fish"  # For fish
source "~/.cargo/env.nu"  # For nushell
source "$HOME/.cargo/env.tcsh"  # For tcsh
. "$HOME/.cargo/env.ps1"        # For pwsh
source "$HOME/.cargo/env.xsh"   # For xonsh
```

- 执行如下命令，使得安装环境启用，并检查版本

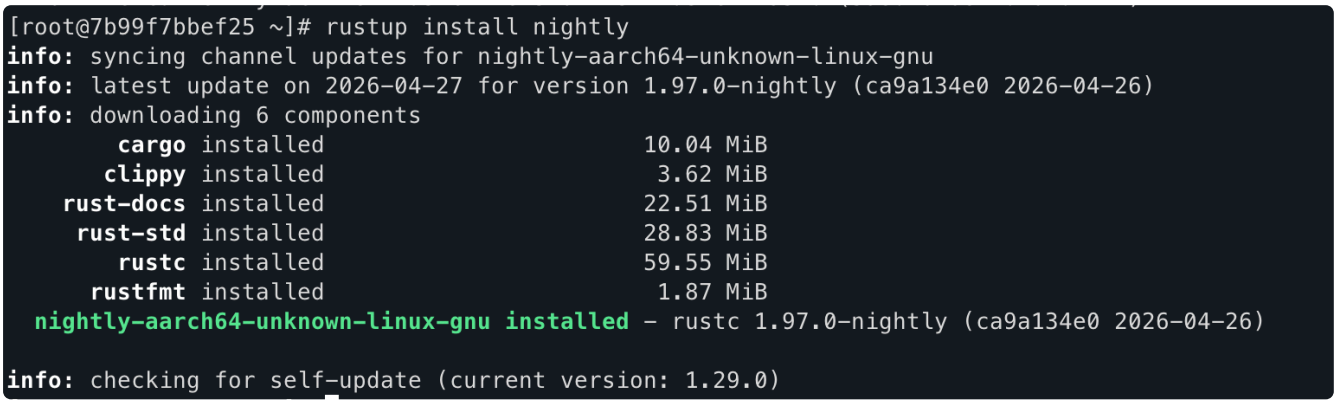
Bash

```
source "$HOME/.cargo/env"
rustc --version
cargo --version
rustup --version
```

```
[root@7b99f7bbef25 ~]# source "$HOME/.cargo/env"
rustc --version
cargo --version
rustup --version
rustc 1.95.0 (59807616e 2026-04-14)
cargo 1.95.0 (f2d3ce0bd 2026-03-21)
rustup 1.29.0 (28d1352db 2026-03-05)
info: This is the version for the rustup toolchain manager, not the rustc compiler.
info: the currently active `rustc` version is `rustc 1.95.0 (59807616e 2026-04-14)`
```

- 安装 nightly Rust 工具链

```bash
rustup install nightly
```

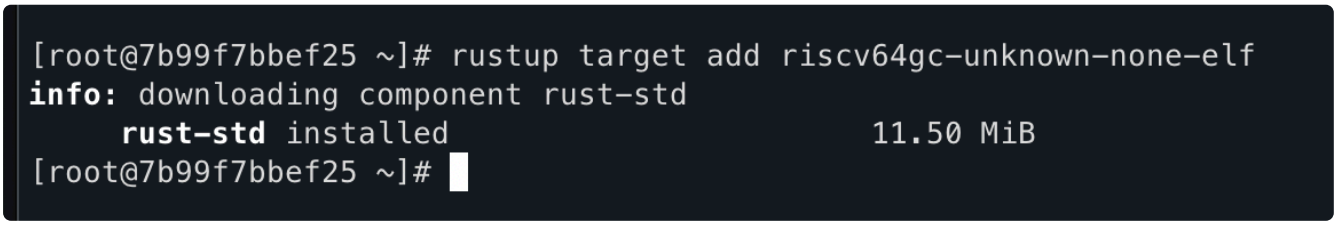


- 把默认 Rust 版本切换成 nightly。

```bash
rustup default nightly
```

- 添加 RISC-V 编译目标

```bash
rustup target add riscv64gc-unknown-none-elf
```



- 安装 `cargo-binutils`

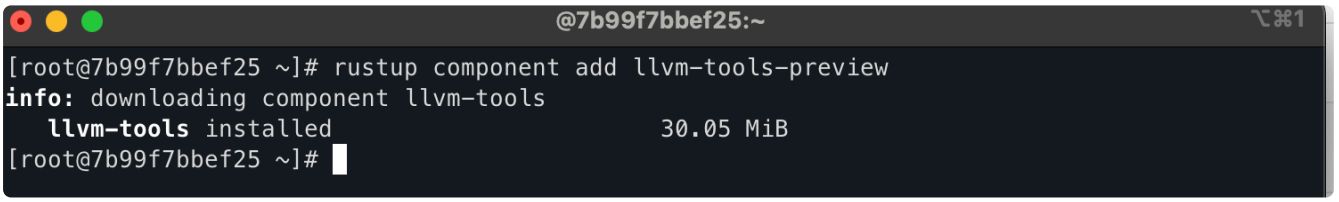
Bash

```
cargo install cargo-binutils
```

- 安装 `llvm-tools-preview` 组件

Bash

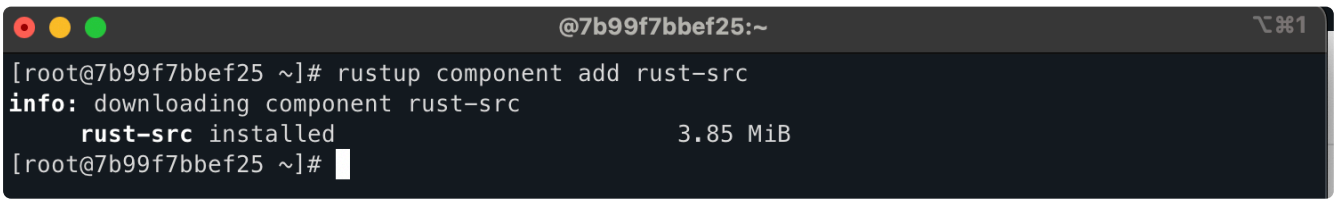
```
rustup component add llvm-tools-preview
```



- 安装 `rust-src` 组件

Bash

```
rustup component add rust-src
```



- 在工作目录下创建一个名为 `rust-toolchain` 的文件，以限制 rust 的版本

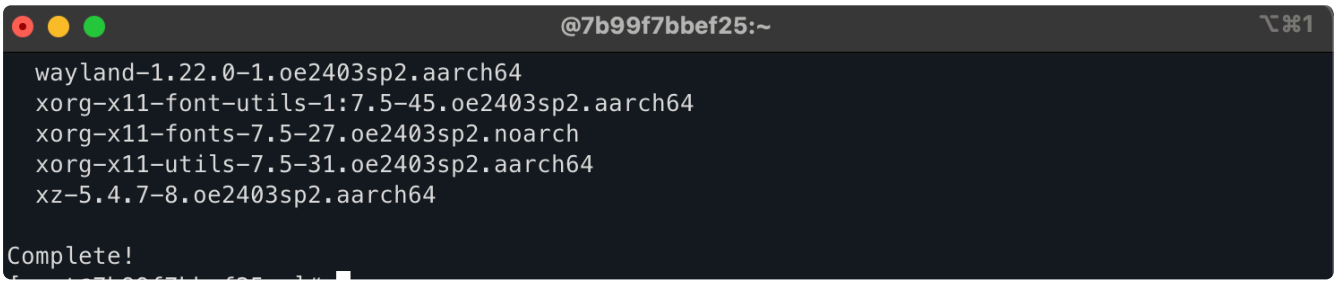
Bash

```
cd /mnt
echo nightly-2022-10-19 > rust-toolchain
```

4.安装 qemu

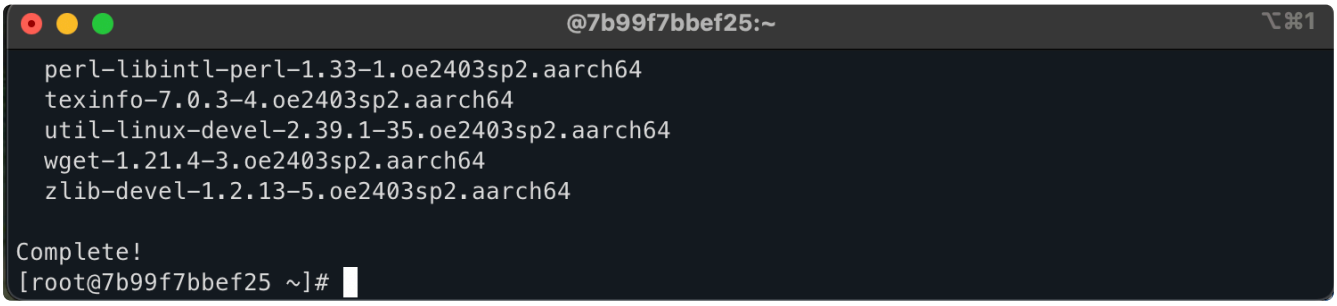
- 安装编译 QEMU 需要的基础开发工具

```bash
cd ~
dnf groupinstall -y "Development Tools"
```



- 安装 QEMU 编译所需的其他依赖

```bash
dnf install -y autoconf automake gcc gcc-c++ kernel-devel curl libmpc-devel mpfr-devel gmp-devel glib2 glib2-devel make cmake gawk bison flex texinfo gperf libtool patchutils bc python3 ninja-build wget xz
```



- 下载 QEMU 5.2 源码

```bash
wget https://download.qemu.org/qemu-5.2.0.tar.xz
```

```
[root@7b99f7bbef25 ~]# wget https://download.qemu.org/qemu-5.2.0.tar.xz
--2026-04-27 09:37:13--  https://download.qemu.org/qemu-5.2.0.tar.xz
Resolving download.qemu.org (download.qemu.org)... 198.18.16.244
Connecting to download.qemu.org (download.qemu.org)|198.18.16.244|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 106902800 (102M) [application/x-tar]
Saving to: 'qemu-5.2.0.tar.xz'

qemu-5.2.0.tar.xz      100%[============================>] 101.95M  11.9MB/s    in 9.5s

2026-04-27 09:37:24 (10.8 MB/s) - 'qemu-5.2.0.tar.xz' saved [106902800/106902800]

[root@7b99f7bbef25 ~]#
```

- 解压 QEMU 源码包并进入

Bash

```
tar xvJf qemu-5.2.0.tar.xz
cd qemu-5.2.0
```

```
@7b99f7bbef25:~/qemu-5.2.0                                    ⌥⌘1
qemu-5.2.0/pc-bios/opensbi-riscv32-generic-fw_dynamic.bin
qemu-5.2.0/pc-bios/edk2-arm-code.fd.bz2
qemu-5.2.0/pc-bios/linuxboot.bin
qemu-5.2.0/configure
qemu-5.2.0/qemu.nsi
qemu-5.2.0/Kconfig
qemu-5.2.0/.gdbinit
[root@7b99f7bbef25 qemu-5.2.0]#
```

- 配置 QEMU 只编译 RISC-V 相关目标并编译安装

Bash

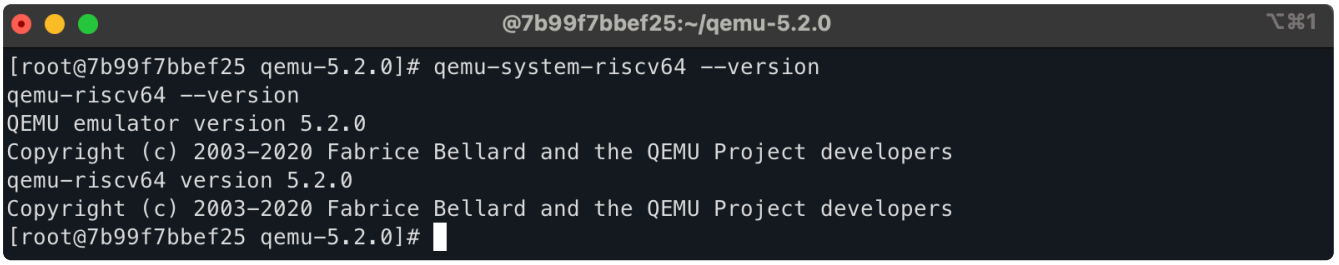
```
./configure --target-list=riscv64-softmmu,riscv64-linux-user && make -j
$(nproc) install
```

```
@7b99f7bbef25:~/qemu-5.2.0                                    ⌥⌘1
e/qemu/firmware
Installing /root/qemu-5.2.0/build/pc-bios/descriptors/60-edk2-aarch64.json to /usr/local/share/qemu
/firmware
Installing /root/qemu-5.2.0/build/pc-bios/descriptors/60-edk2-arm.json to /usr/local/share/qemu/fir
mware
Installing /root/qemu-5.2.0/build/pc-bios/descriptors/60-edk2-i386.json to /usr/local/share/qemu/fi
rmware
Installing /root/qemu-5.2.0/build/pc-bios/descriptors/60-edk2-x86_64.json to /usr/local/share/qemu/
firmware
Installing /root/qemu-5.2.0/pc-bios/keymaps/sl to /usr/local/share/qemu/keymaps
Installing /root/qemu-5.2.0/pc-bios/keymaps/sv to /usr/local/share/qemu/keymaps
make[1]: Leaving directory '/root/qemu-5.2.0/build'
[root@7b99f7bbef25 qemu-5.2.0]#
```

- 验证 QEMU 安装成功

Bash

```
qemu-system-riscv64 --version
qemu-riscv64 --version
```



- 把配置的环境变成自己的 docker 镜像

Bash

```
docker commit -m "configure os experiment environment" -a "Kyy008" gard
eneros myopeneuler
```